

2016.0 RANGE ROVER (LG), 303-03

ENGINE COOLING - TDV6 3.0L DIESEL - GEN 2/TDV6 3.0L DIESEL

COOLING SYSTEM DRAINING AND VACUUM FILLING (G1509365)

GENERAL PROCEDURES

26.10.05	COOLANT - DRAIN, FLUSH AND VACUUM FILL	3000 CC, TDV6	1.8	USED WITHINS
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DRAINING

CAUTION:

Since injury such as scalding could be caused by escaping steam or coolant, make sure the vehicle cooling system is cool prior to carrying out this procedure.

1.

WARNING:

Make sure to support the vehicle with axle stands.

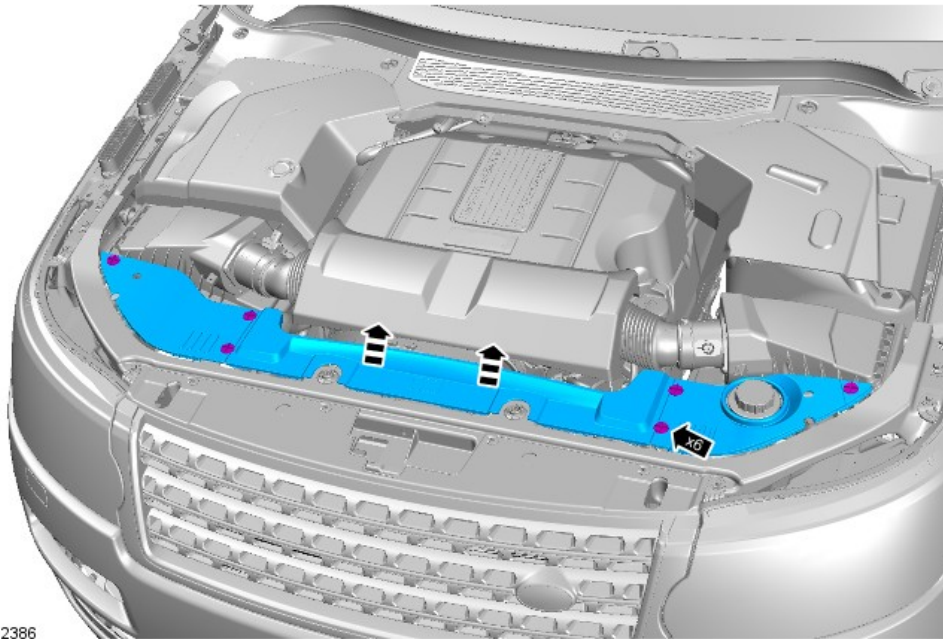
Raise and support the vehicle.

2. Refer to: [Engine Undershield](#) (501-02 Front End Body Panels,

Removal and Installation).

3. Refer to: [Engine Cover - TDV6 3.0L Diesel](#) (501-05 Interior Trim and Ornamentation, Removal and Installation).

4.

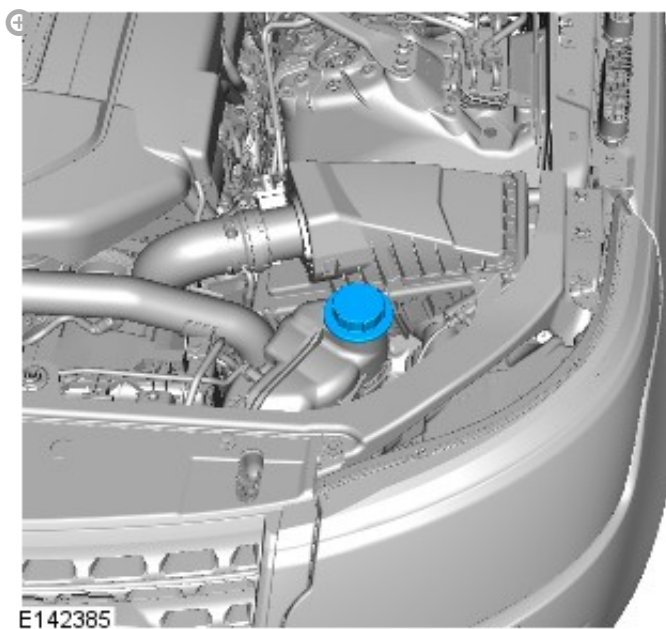


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5.

WARNING:

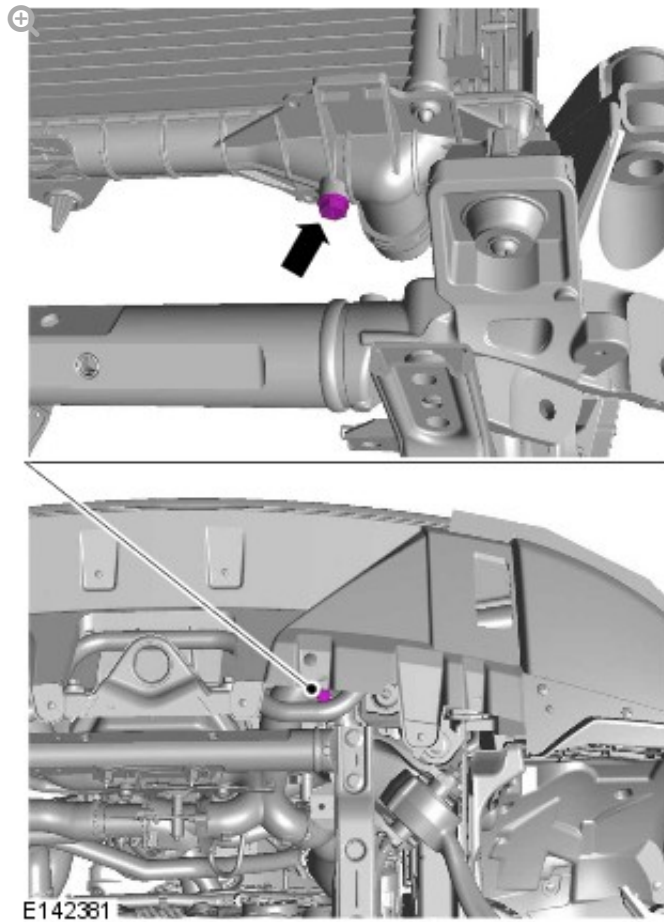
When releasing the cooling system pressure, cover the coolant expansion tank cap with a thick cloth.



6.

CAUTION:

Be prepared to collect escaping coolant.



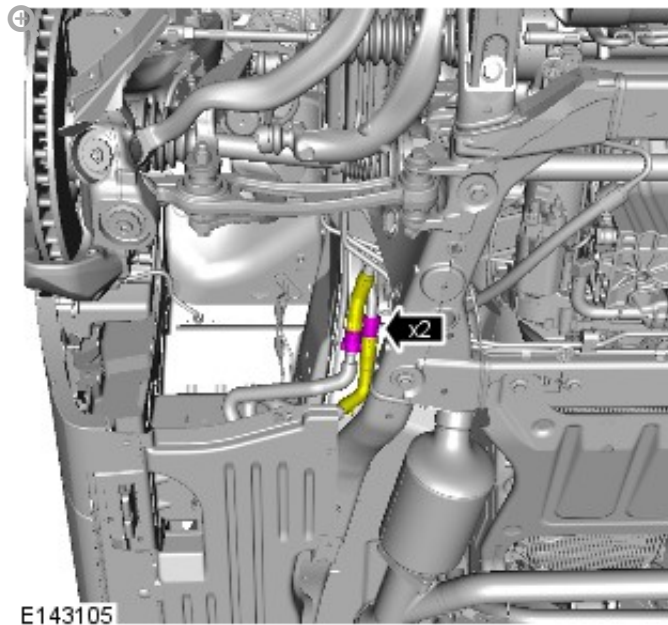
NOTE:

Step 7-10 and 12-13 only needs to be carried out when the auxiliary climate control system has been fitted to the vehicle.

7. Remove the front RH wheel arch liner.
Refer to: [Fender Splash Shield](#) (501-02 Front End Body Panels, Removal and Installation).

8. **CAUTION:**

Be prepared to collect escaping coolant.

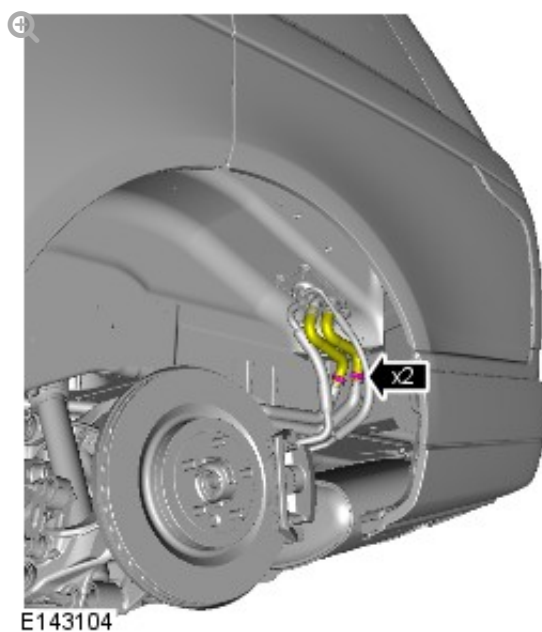


9. Remove the rear LH wheel arch liner.
Refer to: [Rear Wheel Arch Liner](#) (501-08 Exterior Trim and Ornamentation, Removal and Installation).

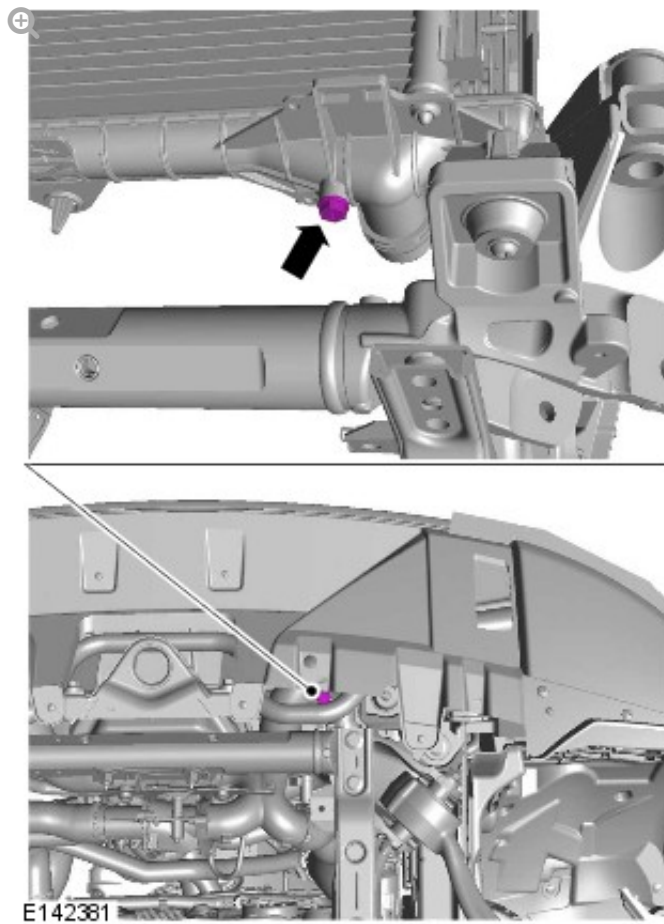
10.

CAUTION:

Be prepared to collect escaping coolant.

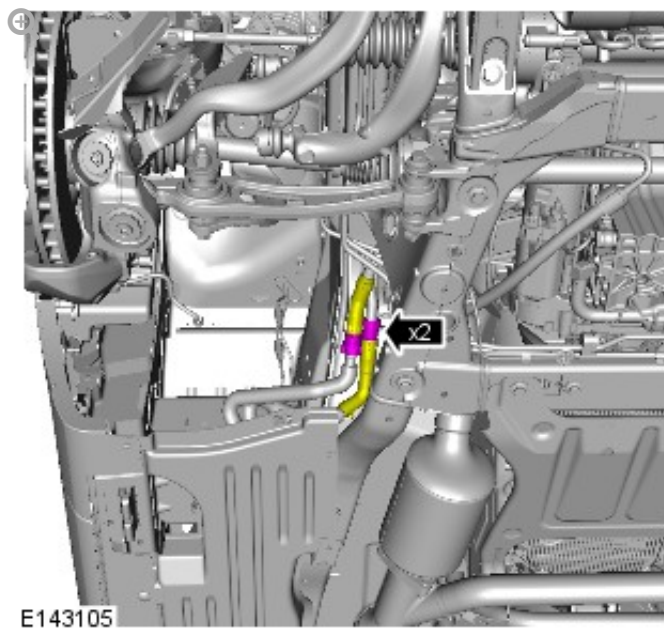


11.

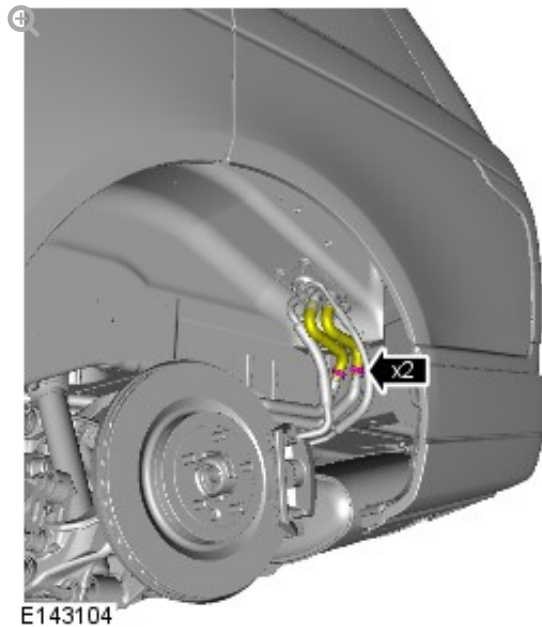


Torque: 5 Nm

12.



13.



14. Carry out the procedure up to step 1 in the filling section three times, filling the cooling system with clean water after the first two drains. At the third refill, use a suitable measuring tool to make sure that the cooling system maintains a 50% mix and vehicles with block heater maintains a 60/40 mix.

NOTE:

Step 7-10 and 12-13 only needs to be carried out when the auxiliary climate control system has been fitted to the vehicle.

FILLING

15.

CAUTIONS:

- Anti-freeze concentration must be maintained at 50%.
- Vehicles with block heater anti-freeze concentration must be maintained at 60%.

1. Install the cooling system vacuum refill adaptor to the expansion tank.
2. Install the vacuum filler gauge to the cooling system vacuum refill adaptor.
3. Install the venturi tube assembly to the vacuum filler gauge.

16.

NOTES:

- Make sure the coolant supply valve is in the closed position on the vacuum filler gauge assembly.
 - The coolant vacuum fill tool needs an air pressure of 6 to 8 bar (87 to 116 psi) to operate correctly.
 - Small diameter or long airlines may restrict airflow to the coolant vacuum fill tool.
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- Connect a regulated compressed air supply to the venturi tube assembly.

17. Position the evacuated air hose into a container.

18. Open the air supply valve.

19.

NOTE:

Make sure the coolant supply hose is positioned into a container of fifty percent mixture of Land Rover Premium Cooling System Fluid or equivalent, meeting Land Rover specification Halvoline XLC and fifty percent water. Make sure no air can enter the coolant supply hose.

Open the coolant supply valve for 2 seconds to prime the coolant supply hose.

20. Apply air pressure progressively until the arrow on the vacuum filler gauge reaches the green segment.
21. Disconnect the compressed air supply line.

22.

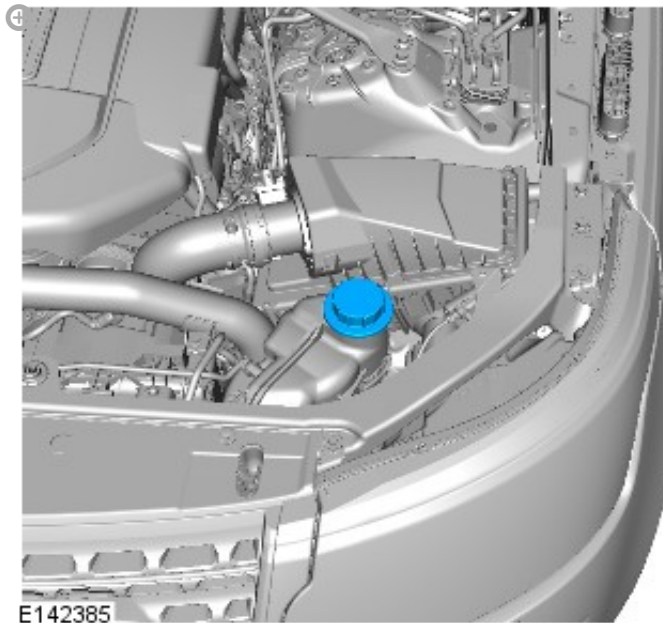
NOTE:

Close the coolant supply valve when the coolant expansion tank MAX mark is reached or coolant movement has ceased.

Open the coolant supply valve and allow the coolant to be drawn into the system.

23. Remove the vacuum filler gauge and cooling system vacuum refill adaptor assembly.
24. Continue to fill the coolant expansion tank to the rim until the coolant level stops dropping.

25.



26. Start and run the engine.

27. Set the heater controls to maximum.
28. Increase the engine speed from idle to 3500 RPM until hot air is emitted from the front and rear face vents.
29. Run the engine until the thermostat opens and hot air is emitted from the front and rear face vents.
30. Switch the engine off and allow to cool.
31. Visually check the engine and cooling system for signs of coolant leakage.

32.

WARNING:

When releasing the cooling system pressure, cover the coolant expansion tank cap with a thick cloth.

CAUTIONS:

- Since injury such as scalding could be caused by escaping steam or coolant, make sure the vehicle cooling system is cool prior to carrying out this procedure.
- Make sure the coolant level remains above the "COLD FILL RANGE" lower level mark.

NOTE:

When the cooling system is warm, the coolant will be approximately 10mm above the upper level mark on the expansion tank with the cap removed.

Check and top-up the coolant if required.